

2022

T40 & T20P Spreading System Repair Training

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01 Product Introduction

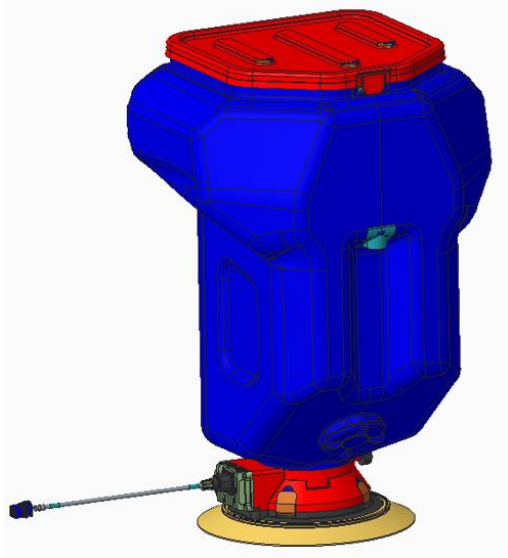
01 Product Introduction



Product Name: T40/T20P Spreading System

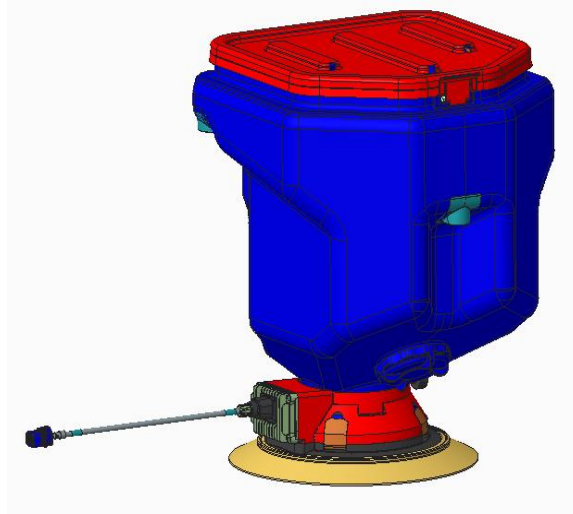
- Improve operating accuracy and reliability.
- DJI increased the load of the T40 spreading system to 50 kg and capacity to 70L, and enlarged the feeding port to facilitate quick feeding.
- T20P Spreading System supports a load of 25 kg and capacity of 35L.
- Using the weight sensor allows the product to control the spreading amount and remaining liquid payload amount in real-time.
- The spreading system supports water washing. With the quick-release structure, it can be quickly switched in three minutes.

01 Product Introduction—Product Comparison



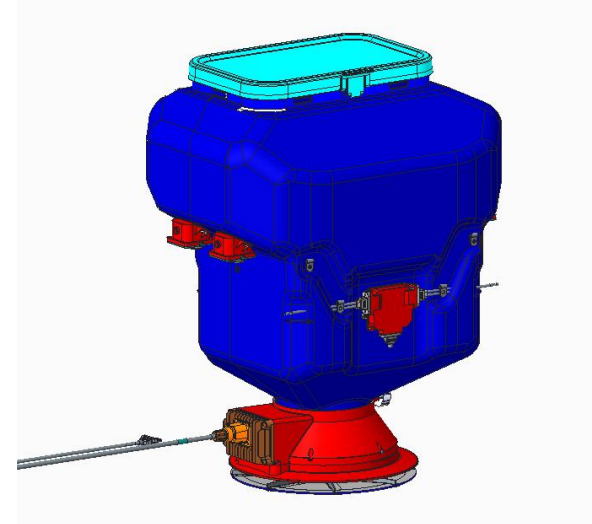
T40 Spreading System

- Monitor the weight of the remaining material in the spread tank in real time.-- The weight sensor is on the aircraft.
- Empty Tank Warning (damping rotation type)
- Quick-Release Design
- Serial Number: 5LM



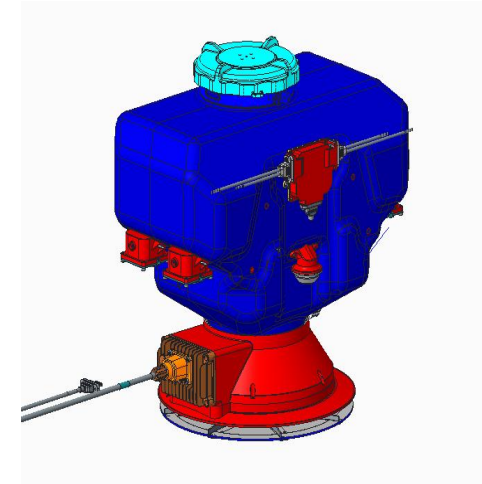
T20P Spreading System

- Monitor the weight of the remaining material in the spread tank in real time.-- The weight sensor is on the aircraft.
- Empty Tank Warning (damping rotation type)
- Quick-Release Design
- Serial Number: 5LN



T30 Spreading System 3.0

- Monitor the weight of the remaining material in the spread tank in real time.
- Empty Tank Warning (damping rotation type)
- Serial Number: 4EP



T10 Spreading System 3.0

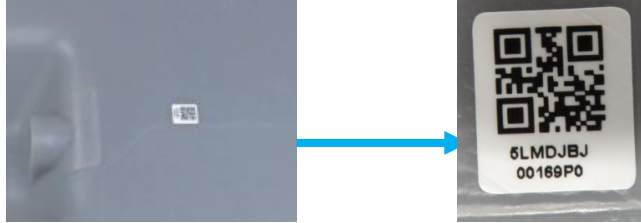
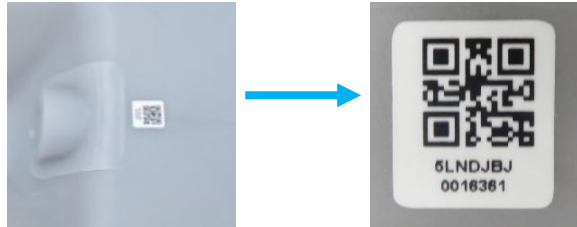
- Monitor the weight of the remaining material in the spread tank in real time.
- Empty Tank Warning (damped rotation type)
- Serial Number: 4EN

Note: The difference between the T40 spreading system and the T20P spreading system is only the size of the spread tank.

01 Product Introduction — Specifications

	T40 Spreading System	T20P Spreading System	T30 Spreading System 3.0	T10 Spreading System 3.0
Compatible Material Diameter	0.5 - 5mm dry solid particles	0.5 - 5mm dry solid particles	0.5 - 5mm dry solid particles	0.5 - 5mm dry solid particles
Spread Tank Volume	70L	35L	40L	12L
Spread Tank Internal Load	50 kg	25 kg	40 kg	10 kg

01 Product Introduction

Material Name	Picture	SN Position and Form
T40 Spreading System		SN Position: On the spread tank Form Sticker 
T20P Spreading System		SN Position: On the spread tank Form Sticker 



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02 Material Identification

02 Material Identification — Main Module



Control Module

Core module of the spreading system.

It is the material available for both T40/T20P spreading system and T30/T10 spreading system.



Servo Module

Controls the change of the hopper gate angle and the opening/closing of the hopper gate.

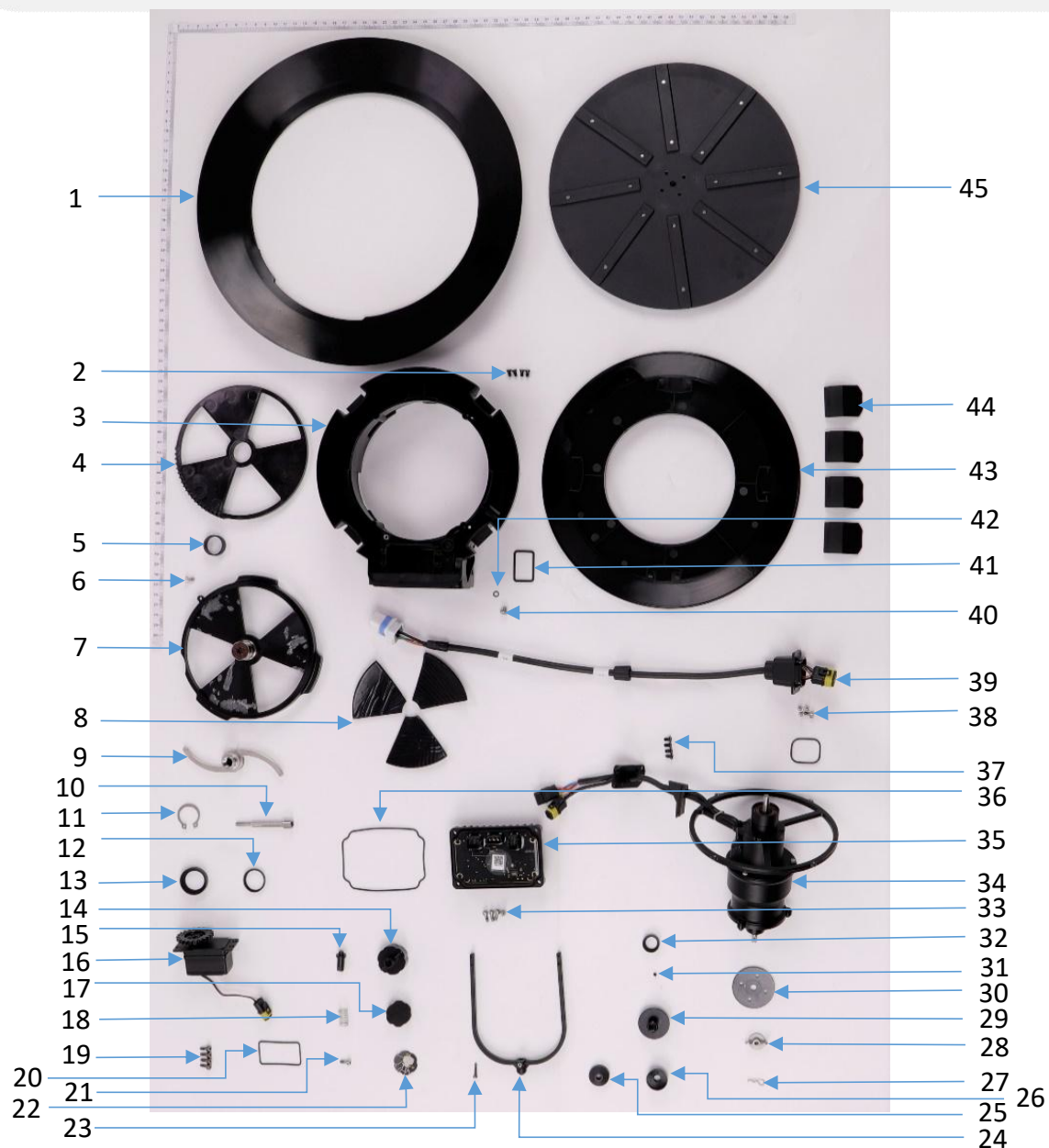
It is the material available for both T40/T20P spreading system and T30/T10 spreading system.



Reduction Gearbox

Includes the propulsion motor and material detection motor: The propulsion motor drives the stirring device to spin, brings the materials to the hopper gate, and spreads them out of the hopper gate, while the material detection motor is responsible for driving the sensing part to rotate and achieve the empty tank warning.

02 Material Identification — Main Module



No.	物料编码	物料名称
1	YC.JG.ZS002376	挡板
2	YC.WJ.LL000096	M30-HB050045-55-05
3	BC.AG.SS000611	甩盘主体
4	YC.JG.ZS002414	大齿轮
5	YC.JG.QX001324	润滑环
6	YC.WJ.L00870	M30-HC060060-55-85 V2
7	BC.AG.SS000608	舱门组件
8	YC.JG.MQ001323	特氟龙贴片
9	BC.AG.SS000607	搅拌棒
10	YC.JG.QX002221	短轴
11	YC.JG.BJ000694	轴用卡簧
12	YC.JG.MY000732	防尘罩2
13	YC.JG.MY000731	防尘罩1
14	YC.SJ.WS001064	旋转锁键
15	YC.WJ.ZZ000968	料箱锁键
16	BC.AG.SS000411	舵机组件
17	YC.BZ.XX000417	锁止机构EVA泡棉
18	YC.WJ.TT000078	锁键弹簧
19	YC.WJ.LL000074	T30-HC080065-60-15
20	YC.JG.MY000341	舵机密封圈
21	YC.WJ.L00868	M30-HC080080-55-85 V2
22	YC.BZ.JJ000329	播撒系统锁止机构贴纸
23	YC.WJ.L00006	杯头螺丝 M2.5*12
24	YC.JG.ZS001366	阻料片
25	YC.JG.MY000387	减速箱防尘罩
26	YC.JG.QX002222	密封套
27	YC.JG.QX002377	R型销手拧螺母
28	YC.JG.QX002377	R型销手拧螺母
29	YC.JG.QX001443	播撒法兰
30	YC.JG.BJ000776	甩盘大垫片
31	YC.WJ.L00005	顶丝3*3
32	YC.JG.MY000733	防尘罩3
33	YC.WJ.L00868	M30-HC080080-55-85 V2
34	BC.AG.SS000606	减速箱组件
35	BC.AG.SS000409	控制模块
36	YC.JG.MY000344	散热盖密封圈
37	YC.WJ.LL000096	M30-HB050045-55-05
38	YC.WJ.L00870	M30-HC060060-55-85 V2
39	YC.XC.XX000847	播撒总信号线
40	YC.ST.LL000112	M40-HC00450039-076020-0003-V
41	YC.JG.MY000723	穿线板密封圈
42	YC.JG.MY000314	排水孔防水圈
43	YC.JG.ZS002356	甩盘上盘
44	YC.JG.ZS002358	甩盘上盘卡扣
45	YC.JG.ZS000276	甩盘

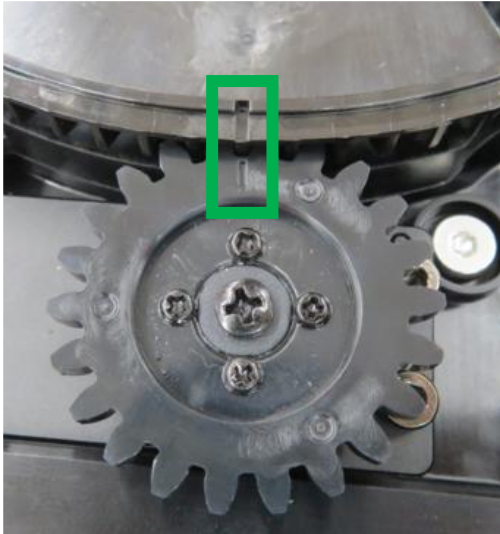




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03 Disassembly and Assembly Notes

03 Disassembly and Assembly Notes



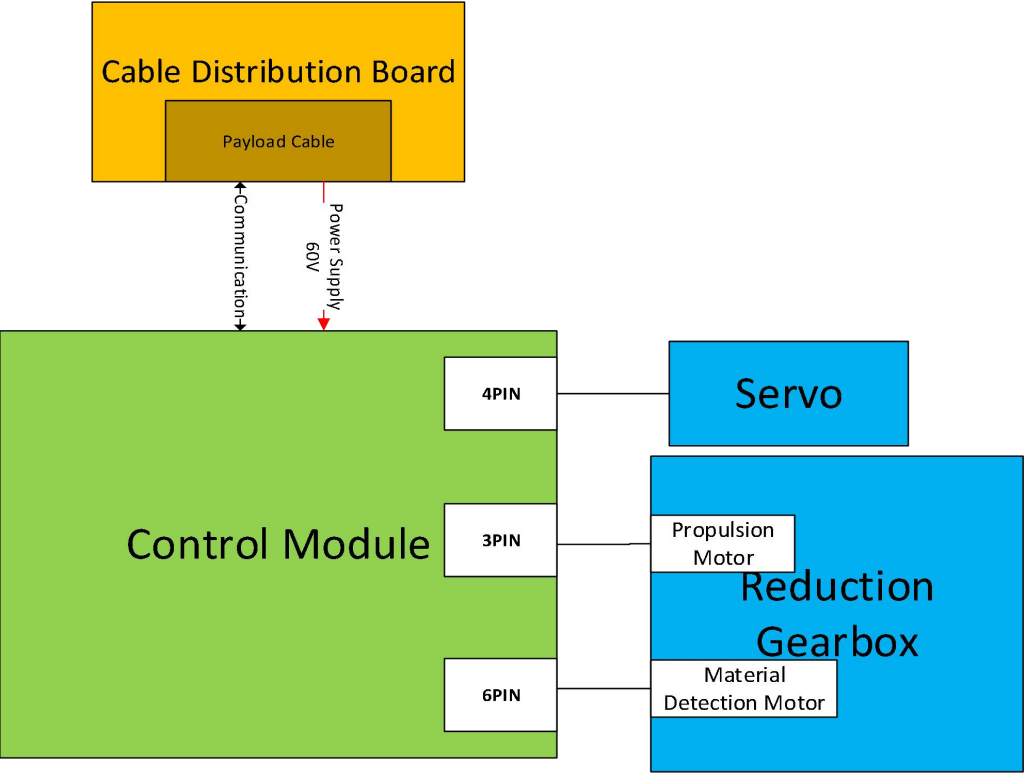
When installing the large gear, make sure that the large gear is aligned with the notch of the small gear on the servo module, and that the large gear can rotate smoothly.



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04 Principles Diagram and Board Introduction

04 Principles Diagram and Board Introduction



Chip	Function
Control Module	Core module of the spreader; collects ESC signals, and servo signals, and controls the various resources of the entire spreader.
Servo	Controls the change of the hopper gate angle and the opening/closing of the hopper gate.
Reduction Gearbox	Drives the stirring device to spin, brings the materials to the hopper gate, and spreads them out of the hopper gate. There are two built-in motors. The material detection motor drives the structure attached above the motor to rotate. The control module determine whether to make an app prompt for an empty tank based on the current value of the motor; the propulsion motor drives the spinner to rotate and spreads the materials that fall on the spinner disk.



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05 Test Plan and Fixture

05 Test Plan and Fixture

Product	Content	Fixture/Device	Software (Name/Version)	Damage Assessment Required	Repair Required
Spreader	Firmware Package Update	None	DJI Assistant 2	No	Yes
	App Functionality Test	None	Built-in App	Yes	Yes

05 Test Plan (Tests after components replacement)

Product	Test Item	Control Module	Reduction Gearbox	Servo	Cable
Spreading System	Firmware Package Update	√	√	√	√
	App Functionality Test	√	√	√	√

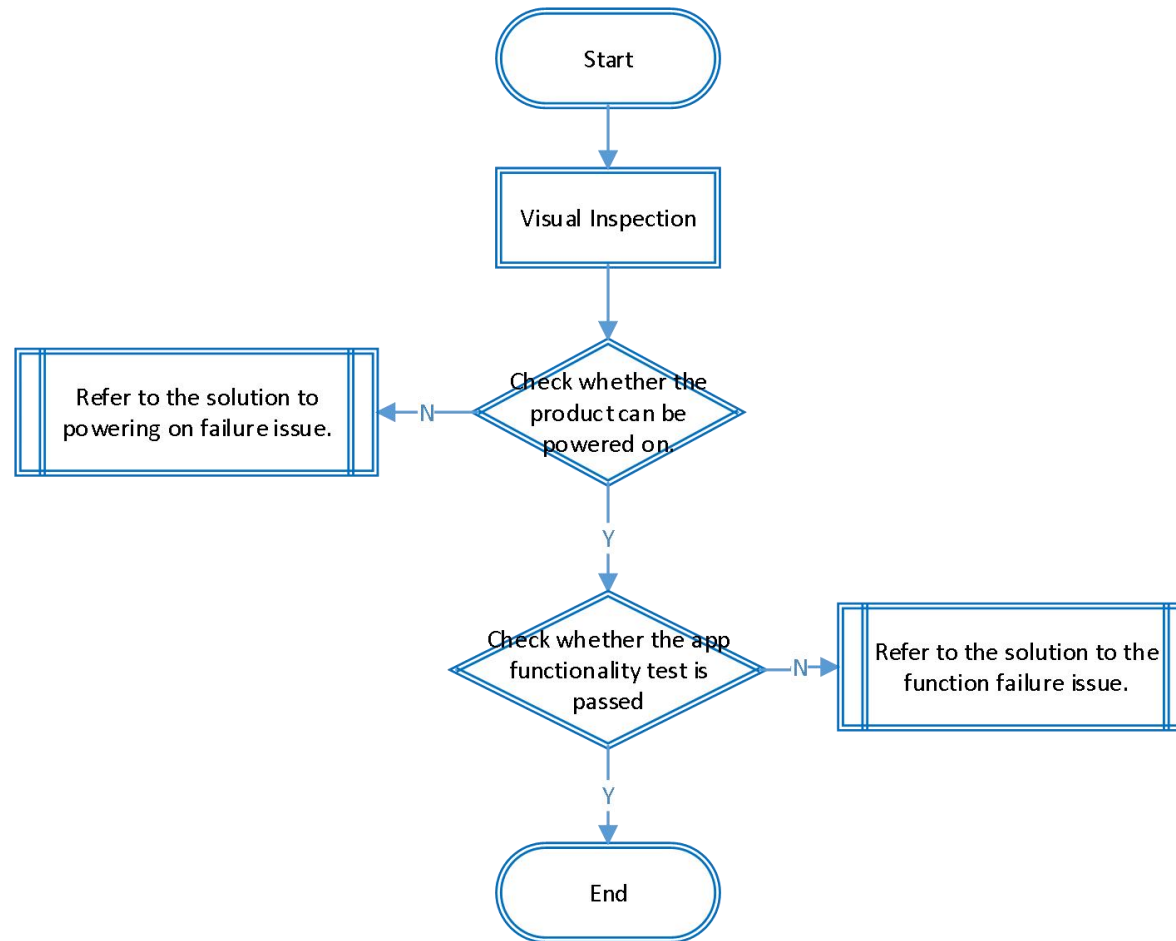


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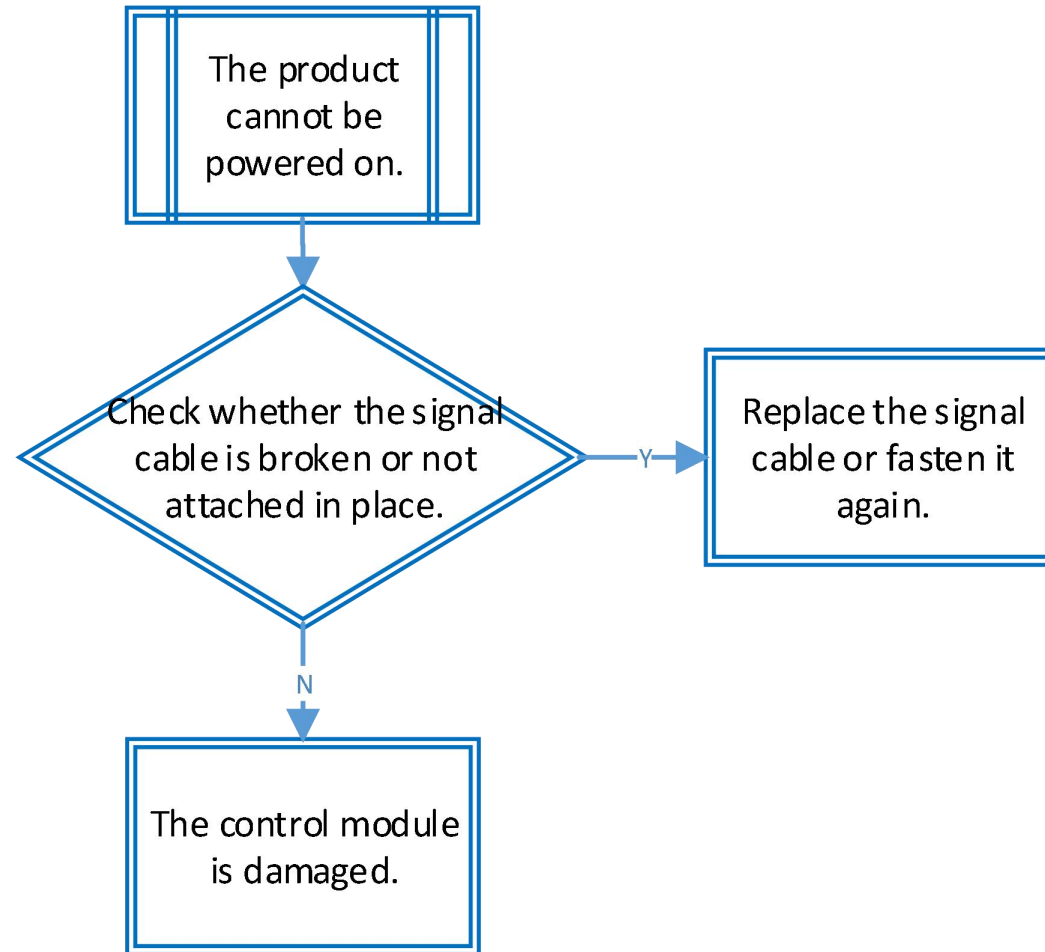
06 Malfunction Assessment and Analysis

06 Malfunction Assessment and Analysis — Common Issues

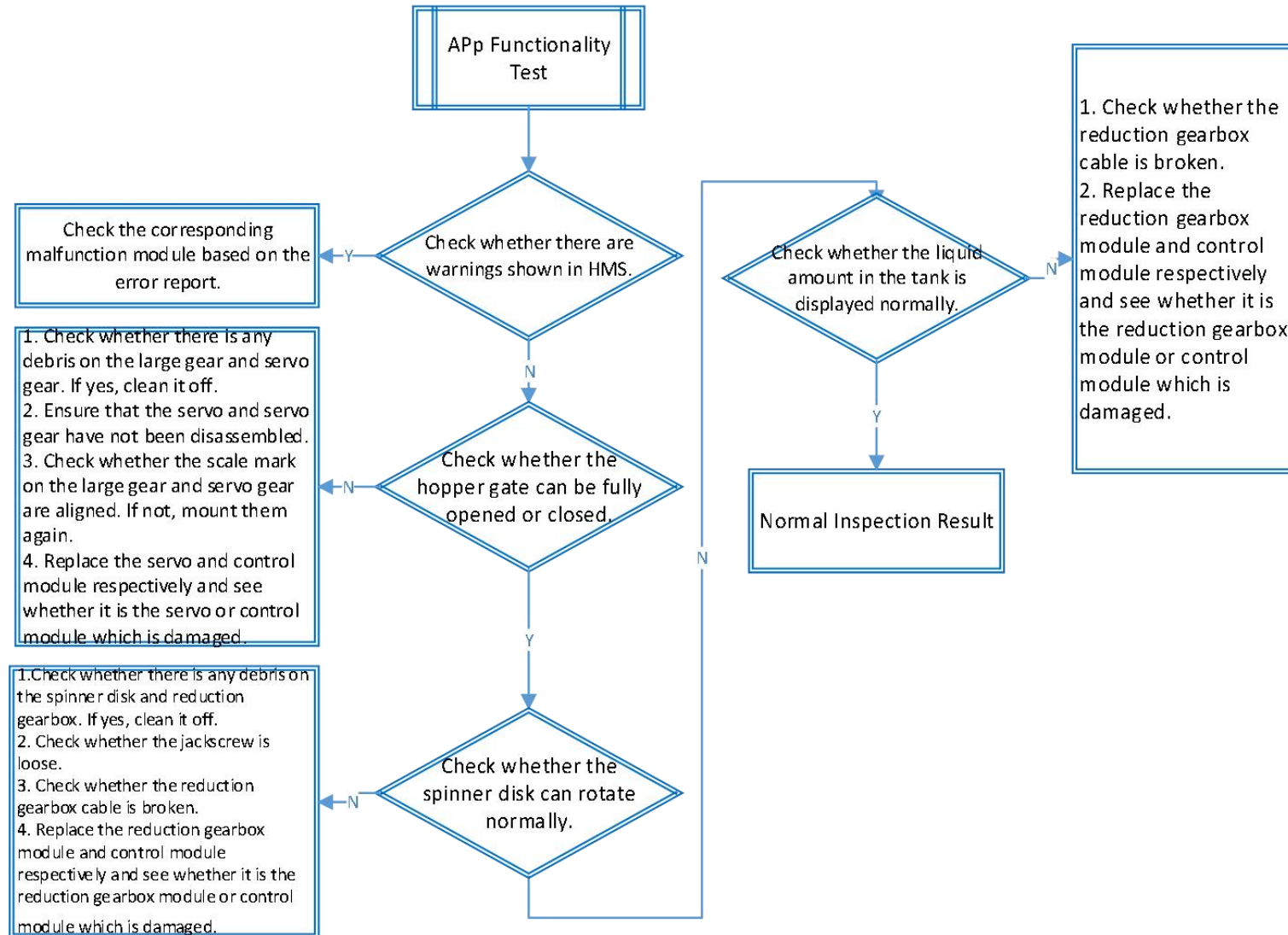
Perform damage assessment by using the app.



06 Malfunction Assessment and Analysis — Common Issues



06 Malfunction Assessment and Analysis — Common Issues



06 Malfunction Assessment and Analysis — Water damage

Check whether the liquid contact indicator has turned red and whether the board is moldy.

Liquid Contact Indicator		Board Phenomenon	Assessment
Has a liquid contact indicator or not?	Liquid Contact Indicator Status		
With a liquid damage indicator	Turned red	Corrosion/water stain/water mark	Determine the product is water damaged.
		No Corrosion/water stain/water mark	Determine the product is not water damaged.
	Color unchanged/unbleached	Corrosion/water stain/water mark	Determine the product is water damaged.
		No Corrosion/water stain/water mark	Determine the product is not water damaged.
No liquid damage indicator	NA	Corrosion/water stain/water mark	Determine the product is water damaged.
		No Corrosion/water stain/water mark	Determine the product is not water damaged.



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FAQ



Thank you for watching!